

Arun V. Eluvathingal
Department of Electrical Engineering
IIT Madras, Chennai 600036, India

April 14, 2026

Editor-in-Chief
IEEE Transactions on Power Delivery

Dear Editor,

We submit for your consideration the manuscript:

**Hardware-in-the-Loop Firmware Validation of the SEL-411L
Advanced Line Differential Relay via SAMBPS and IEC 61850 Sampled Values**

to be considered for publication in *IEEE Transactions on Power Delivery*.

Context. Commercial protection IEDs are typically validated against vendor test sheets under idealised, synchronous-generator-dominated conditions. As inverter-based resources (IBR) increase their penetration, the adequacy of such validation is no longer assured. This paper presents the first systematic, HIL-based firmware validation of the SEL-411L line differential relay under an IBR-stressed grid, executed through the Self-Adaptive Model-Based Protection System (SAMBPS).

Key contributions.

1. **SAMBPS co-processor architecture (Proposition 1).** A five-stage critical-path analysis proves that the combined SEL-411L / SEL RTAC 3530 pipeline meets the 25 ms protection coordination window: $t_{\text{total}}^{\text{max}} = 19.5$ ms.
2. **Float32 κ_n bound (Proposition 2).** Using Wilkinson/Wedin perturbation theory and Higham's floating-point arithmetic, we prove $|\delta\kappa_n| \leq 0.148$ at $\kappa_n = 30$, yielding a safe trip margin of $29.85 \ll 30$.
3. **48-scenario HIL validation.** Three-phase, line-to-line, single-line-to-ground, CT saturation, channel loss, PV IBR ($k_{\text{ibr}}=1$), and SG external fault scenarios are tested; 46/48 pass (95.8%).
4. **Two documented deviations.** D3 (IBR LL fault: firmware confirm-timer minimum) and D4 (N-1 simultaneous: +0.3 ms GOOSE latency) are characterised and both resolved without firmware modification.
5. **Trip-latency benchmark.** SEL-411L latencies ($t_{50} \in [19.1, 22.7]$ ms) align within ± 2.6 ms of the companion SEL-300G HIL result (TR-67), demonstrating cross-device reproducibility of the SAMBPS framework.

Relationship to companion papers. This work is the G1 gap-paper in the SAMBPS series. Algorithm development appears in companion manuscripts submitted to *IEEE Transactions on Power Delivery* (J1: line differential 87L) and *IEEE Transactions on Smart Grid* (J4: unified five-layer architecture). The present paper focuses exclusively on firmware-level hardware validation and numerical precision analysis of the commercial relay platform.

Suggested reviewers.

- Prof. Ali Hooshyar, University of Toronto (IED validation, IBR protection)
- Prof. Geza Joos, McGill University (power electronics, IEC 61850 integration)
- Prof. Wilsun Xu, University of Alberta (harmonic/IBR phenomena, power quality)
- Dr. Evangelos Farantatos, EPRI (protection systems, hardware validation)

The manuscript has not been submitted elsewhere and all authors have approved the final version.
The work is original and no figure, table, or section reproduces previously published material.

Sincerely,

Arun V. Eluvathingal
Department of Electrical Engineering
IIT Madras, Chennai 600036, India
arun.eluvathingal@iitm.ac.in